

The Future of Motherboards



Motherboards now have Serial ATA support, Dolby Digital sound and RAID controllers onboard. Support for Dual Channel DDR, connectors for card readers, IR ports and onboard LED indicators for fault diagnostics are some other futuristic features that have not made much inroads into mainstream motherboards.

Chipset: New processors need newer chipsets to support them. Some of these are the 875P chipset from Intel, nForce2 from nVidia, UniChrome KM400 from VIA (for the AMD processor), SiS655 and SiS748-Dual channel DDR333 (for the P4), and DDR400 with support for AGP8X (for the AMD). These are, however, found only in high-end boards.

There are some very recent launches for the AMD family—nForce3 Pro 150, 250 and Intel has released the I875P (earlier known as Canterwood) and Springdale chipsets. nForce3 Pro is build specifically for single processor workstations. nVidia has bundled the latest goodies with nForce3 Pro's single chip design such as Serial ATA/ATA-133 with RAID.

nForce3 Pro is to be a single chip design, making the manufacturing and designing of boards easy for third-party manufacturers. It's built using 0.15 micron fabrication process, allowing room for more features. The single chip will most likely make over-heating less of a problem, and reduce latency between the motherboard sub-systems, which will translate to increased platform performance.

The I875P chipset will support DDR-400 memory and with quad-pumped 800 MHz FSB and the new HyperThreaded P4 processors. The Canterwood chipset has the tweaked 'turbo mode' PAT (Performance Acceleration Technology), named after CEO Pat Gelsinger. The Springdale chipset, on the other hand, will be available in three flavours—865G, 865P and 865PE.

The year 2004 might see the replacement of Canterwood and Springdale by the Grantsdale chipset. This will sport the cost-efficient LGA (Land Grid Array) 775 pin layout which will oust the current BGA (Ball Grid Array) and PGA (Pin Grid Array) pin layout. There's going to be extensive replacement of current generation chipsets across the board, and support for new technology as well. But all this is targeted mainly at the high-end segment, and will take some time to percolate to the majority of home users.

RAM Support: Intel still continues to offer support for RDRAM by releasing its last RDRAM chipset (the 850e), but there's zero support from other chip manufacturers. DDR memory is what's being pushed in a big way by major mainboard chipset manufacturers. Gigabyte has already released its board based on SiS655, supporting Dual Channel DDR memory. Memory supporting dual-channel DDR is pretty expensive at the moment, because of the limited availability. But as time passes, prices will fall to realistic levels. The older SDRAM memory is still around but will diminish soon.

Manufacturers such as IBM, Motorola and Infineon are

specifically involved in the research of a new form of solid-state memory called Magnetoresistive RAM (MRAM). This type of RAM uses the principle of alignment of magnetic particles as the basis of storing zeroes and ones in a digital system.

An inherent advantage with MRAM is that it since it's based on the principle of magnetism; it retains whatever information is contained in it even after power is switched off. Additionally, these elements do not need an electric charge to retain information, hence power consumption of these devices is minimal compared to that of DRAM. Finally, with the absence of the delays associated with transferring electricity through the storage elements for retaining the information, MRAM is expected to be up to 30 times faster than DRAM.

MRAM is thus poised to give rise to a new breed of instantly-on computing devices.

The technology is already up and running in some high-tech labs around the world. But only after the fabrication process for this memory becomes commercially viable will it make its way into mainstream home PCs and handhelds.

Graphics support: The nForce chipset was the first in recent times to deliver a decent on-board graphic solution. As newer games and ever-demanding software are released, users are always looking for that extra bit of graphical processing power. The AGP 8X graphics interface has already become the standard interface in most boards today.

Interfaces: Support for up to 10 USB 2.0 ports, IEEE 1394 (FireWire), card readers etc., are some features that will soon become standard with every motherboard. As users acquire more and more devices with PC connectivity—such as digital cameras and cell phones—it will become necessary to have such features on board. PCI-X, which has a 1 GB transfer rate, has a performance boost of at least 32 times over the current PCI standard. PCI-X standard is backward-compatible and therefore will support PCI cards as well.

DPS: DPS (dual power system) is a new technology that enhances the stability of the system. This technique allows the board to have a total of six phases for power circuitry—twice the number recommended in the standard motherboard design guideline. A 3-phase power circuit Dual Power Voltage Regulator Module (DPVRM) daughter card can be added on the board, to help provide more stable power circuits.

Design: Motherboards are likely to go both bigger and smaller. With additional features going into standard motherboards, it might get bigger rather than smaller. However, single chip designs and high level of integration will drive sizes down as well. Companies as ViA have come out with a complete platform known as IDEN, which has an onboard C3 processor running at 1 GHz, 5.1 channel sound and graphics. The board is right now targeted at basic office users and comes in a very small form factor.

